SAMPLE

Worlds Beyond Earth: Dwarf Planets

Ceres, Eris, Haumea, and Makemake

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Introduction

There is a shelf in the Solar System's library where the misfit worlds are kept. It holds a queen of the asteroid belt with buried brine, a heart-marked king of the deep cold, an egg-shaped spinner with a ring, a world heavy enough to break a century of textbooks, and a growing crowd of candidates in the outer dark, each one round, each one orbiting the Sun, and none of them counted as a planet. They are the dwarf planets, and this book is their story: what they are, why the category exists, who the members are, and why the argument over their names is really an argument about what a planet means.

The story begins with a discovery that broke the Solar System's furniture. For most of the twentieth century the roster was stable: nine planets, the ninth a small oddball beyond Neptune called Pluto, discovered in 1930 and grandfathered into the family on a technicality of timing. Then the outer system began to give up its secrets, the Kuiper belt was found, the small worlds beyond Neptune came in by the hundreds, and in 2005 a team in California announced a discovery bigger than Pluto itself. The astronomy community faced a question it had dodged for seventy years: if Pluto is a planet, what about the new world, and the next one, and the fifty after that. Either the Solar System had dozens of planets, or Pluto had never been one.

The astronomers chose, at a famous and fractious meeting in 2006, to define the word planet for the first time,

Note also the category's one unusual legal feature, because it explains some of the paperwork the census chapter describes: the definition was written for the Solar System alone. The clause about orbiting the Sun, written before the exoplanet census matured, leaves the thousands of known planets of other stars formally outside the union's definition, a gap the community bridges by usage, calling the exoplanets planets in every paper and poster, and the paperwork's echo is enjoyable: the word that was defined to end one argument quietly excluded the biggest population it could have covered, and the union's committee, to its credit, has left the bridge to common sense. And the definition, built in a hurry and argued about ever since, created the category this book describes. A dwarf planet, by the ruling, is a body that orbits the Sun, is massive enough to have been pulled round by its own gravity, but has not swept its orbital neighborhood clean of its rivals. The category is a compromise between geology and dynamics, and the reader will meet both sides of the argument in these pages, because the argument has never really ended: to the planetary geologists, these worlds are planets, full stop, and to the dynamicists, the neighborhood matters more than the shape.

The cast, in decreasing size, is a good walk through the modern Solar System. Eris, the heavyweight whose discovery started the whole affair.

End the introduction with the candidate count's honesty, so the reader's map has its scale lines drawn. The nine that head this book's roster are the confident tier, the class's certified members by the community's working consensus. The borderland behind them runs perhaps a dozen deep, suspects whose rounds are argued. The predicted population runs to the hundreds, mostly small, mostly dark, the models' arithmetic. And draw the map's last line: the boundary between the tiers is an instrument's reach, every better telescope moves a world inward from predicted to candidate to member, and the roster the reader holds is the first page of a census that will outlive everyone who ever printed it. Pluto, the demoted ninth planet and the best explored of the class. Haumea, spinning every four hours into an egg. Makemake, Gonggong, Quaoar, Orcus, the outer system's middle class. Sedna, the deep-space wanderer with the strangest orbit known. And Ceres, the innermost member, a world of brine and salt in the asteroid belt, visited by a spacecraft that watched its bright spots gleam. Nine names the astronomers broadly agree on, a borderland of candidates beyond them, and a hundred or more expected in the Kuiper belt's cold inventory, waiting for the counts.

This book describes them honestly:

A note on what the book is not, so the reader's expectations are set kindly: it is not a polemic. The author has watched the naming argument from the inside for twenty years, holds a mild opinion the reader will detect in the later chapter's prose, and has written the chapters to be fair to both the ruling and the dissent, because both sides contain the field's best people arguing about something that matters to them. Whoever wants a verdict will find the book's honest one in its final chapter, and whoever wants a fight can supply their own; the worlds themselves, the author would like to note, have been entirely indifferent to the proceedings, and the book takes their cue. What the spacecraft actually saw at Ceres and Pluto, what the telescopes have measured of the rest, what remains unknown, and why the unknowns are the interesting part. The reader will finish knowing the roster, the rules, the argument, and the worlds, and can hold a private opinion about the naming, which the author hopes will be a mild

one.

Chapter 01: Drawing the Line: The Definition and Its Three Clauses

Astronomy had a planet problem for a very long time, and the problem, in its simplest form, was that nobody had ever defined the word.

The history's earlier dress rehearsals deserve one paragraph, because 2006 was not the field's first census crisis. The nineteenth century's asteroid discoveries forced the first: Ceres and its neighbors entered the roster as planets, stayed for decades in the textbooks' lists, and were eased out by the census's own arithmetic, as the belt's population made the class unlistable, a reclassification nobody now disputes and nobody at the time enjoyed. The asteroid precedent holds the 2006 story's template: the community has faced an open-ended population before, drew the same line between the majors and the crowd, and the nineteenth's demoted class, as it turned out, was the Solar System's most scientifically important small bodies, a precedent the twenty-first's demoted class has already begun to repeat. The ancients called planets the wanderers, the lights that moved against the stars, and the modern era inherited the word as a name for whatever the community agreed to list, a definition by census rather than by rule. The census served until the outer system broke it, and this chapter is about the rule the astronomers wrote when the census failed, and about what the rule says, clause by clause, because each clause carries a philosophy.

The first clause is the oldest: a planet orbits the Sun. The wording, written in 2006, quietly evicts the exoplanets from the definition, an accident of history the astronomers have lived with since, and it also defines the boundary of the subject: everything in this book is a denizen of our own system, whatever the census of other systems may someday say. The clause sounds trivial and is not, because it draws the line against the moons. A moon can be as large as a planet and as geologically alive, the largest moon of the Solar System is bigger than Mercury, and the line's arbitrariness deserves noting early: the definition is about address, not about nature.

The second clause is the geologist's clause, and it is the one this book's subjects satisfy most proudly:

Put a sentence behind the clause's numbers, because the reader will want the scale. For a rocky body, roundness arrives somewhere around six hundred kilometers across and up, the gravity of a body that size sufficing, over geological time, to flow its own material into a sphere; for an icy body, the threshold sits lower, the ice flowing at smaller sizes, perhaps four hundred kilometers, which is why the outer system's candidate list reaches down into sizes the belt's rocky bodies could never round. Hold the two numbers as the category's physics: round is a mass statement, ice rounds early, rock rounds late, and the borderland's arguments, the reader will see, are mostly arguments about which number applies to which suspect. A planet-like body must be massive enough that its own gravity pulls it into a rounded shape, the condition the literature calls hydrostatic equilibrium. Below a certain mass, a body keeps whatever lumpy shape its collisions gave it; above it, the rock and ice flow, over millions of years, into a sphere, and the world becomes round, which is the visible signature of a body that has been geologically real: it has had an interior, a warmth, a history of flowing and settling. The threshold depends on composition, colder icy worlds round themselves at smaller sizes than rocky ones, which is why the outer system's dwarfs are smaller than the rocky planets' moons, and the clause's fuzzy edge, how round is round, is one of the field's running arguments, and the reason the candidate list has a borderland rather than a border.

The third clause is the dynamist's clause, and it is the one that ends the argument in the ruling's favor: a full planet has cleared its orbital neighborhood, swept or ejected the other bodies near its orbit, so that it dominates its zone rather than sharing it. The eight planets have done this; Pluto, embedded in the Kuiper belt among thousands of neighbors, has not, and neither have any of the subjects of this book. The clause is quantitative at heart, the distinction between a whale and the biggest fish in a school, and its honesty bears understanding: it is a statement about dynamical dominance, about what a body has done to its surroundings, and it is also the clause the geologists cannot accept, because it grades a world by its address and not its nature, a standard under which the Earth, moved to the Kuiper belt, would fail.

The ruling's consequence, stated plainly, is the category of this book:

A footnote from the ruling's aftermath deserves recording, because the committees kept working. Two years after the vote, the union added a subcategory for the class's trans-Neptunian members, coining a word, plutoid, to gather the outer dwarfs under one label, a term the specialists use and the public has thoroughly ignored, and the footnote's lesson about the naming's sociology: the official vocabulary keeps refining itself in the background, the public's vocabulary froze at the vote, and both are correct usages in their own rooms. A body that meets the first

two clauses and fails the third is a dwarf planet, a word the meeting minted on the spot. The category is, be aware, a legal fiction of the best kind, a definition that lets the community keep counting planets at eight, and the chapter's honest summary is the one the argument in a later chapter will expand: the Solar System's small round worlds are, by nature, planets to the geologists and dwarfs to the dynamicists, and both descriptions are correct within their own philosophy.

The third clause's biography gets a paragraph. The neighborhood-clearing clause is the definition's only dynamical test, and its history reaches back to the belt's founding chaos, the young Jupiter having scattered the zone between it and Mars, the account keeping the clause's reach, the same physics that emptied Ceres's zone populating the reservoirs beyond Neptune, the definition file closing on the standing link, the clause the astronomers wrote in 2006 describes an event that happened in the system's first ten million years, and the dwarf planets are its survivors.

The resolution's text gets a line. The 2006 resolution defines a planet by three tests and names the class that fails the third alone, the account keeping the text's brevity, the whole definition being three sentences long and the argument about it having filled a library, the definition file closing on the standing proportion, the shortest ruling in planetary science carries the longest dissent.